

Olist E-Commerce Analysis: Case Study

Situation:

Olist is Brazil's largest E-commerce marketplace, connecting small and medium-sized sellers to major platforms like Mercado Livre. Instead of selling directly to customers, Olist acts as a market aggregator – where they charge a monthly commission fee to sellers and handle the logistics for them. Because of this Olist's revenue is tied directly to their seller performance and customer satisfaction.

Brazil has a very challenging E-commerce environment for multiple reasons. Firstly the environment is very difficult with underdeveloped logistical infrastructure making delivery difficult. This is especially true for the northern regions. Another reason for the challenge is that customers rely on bank slips along with credit cards for payments so Olist has to work with multiple forms of payments.

This analysis examines 100,000 real orders on Olist from 2017-2018 which was a period of rapid growth for the marketplace. Using SQL for data extraction and Tableau for visualization, the goal is to find what are the main drivers of revenue, which areas customer satisfaction is breaking down and what Olist can do to sustain and improve growth.

Findings:

1. From January 2017 to mid 2018 revenue grew 23x with \$15.4 million in total revenue from all of the orders. Peak revenue occurred of \$1,153,528 in November 2017 due to the black Friday sale dwarfing the January 2017 revenue of only \$127,545. By 2018 revenue plateaued to about 1 million per month suggesting a growth ceiling. Across this period the average order value was \$159.86, with credit transactions accounting for about 74% of revenue. Credit Card transactions also had a much Average Order Value (AOV) of \$162 compared to an AOV of \$62 for voucher based purchases indicating that discounts may attract lower value customers.

2. Based on the Olist RFM segment findings the overall customer base is split between a massive gone group, a tiny champions group and smaller intermediate groups. Starting with the gone group 58% of Olist customers with completed orders from January 2017 to mid 2018 are classified as lost, while only 316 customers, 0.3% of overall customers are classified as champions. Regarding the intermediate groups the needs attention group has roughly 14,207 customers with an average spending of \$333 per customer. If Olist is able to get each of these customers to purchase even one more time, Olist is then looking at an additional \$4.7 million in revenue. Potential loyalists and recent customers are sitting at 12,273 and 9,335 respectively.

3. A cohort is a group of customers who make a purchase in the same month and cohort retention tracks what percentage of a cohort comes back to buy again after their first month of purchase. The cohort retention for Olist never exceeds 0.7% in any month. This pattern held consistently between January 2017 and mid 2018 even with the November 2017 black Friday cohort of 7,060 customers. Meanwhile a healthy cohort retention rate for an E-commerce marketplace is anywhere between 20%-30%. This cohort retention problem for Olist never improved over time.

4. For categorial performances health and beauty took the top spot generating \$1.23 million in revenue. Watches and gifts took the second spot with \$1.16 million in revenue and they also had the highest AOV at \$199. There were also 4 categories all rated below 4 stars representing \$2.9 million at risk revenue. Those categories were Bed Bath & Table at \$1 million in revenue and 3.92 star rating, Computers & Accessories at \$892K in revenue and 3.98 star rating, Furniture Decor at \$712K in revenue and 3.95 star rating, and Office furniture at \$266k in revenue and 3.52 star rating.

5. There was a strong correlation between delivery times and reviews. For example delivery times from 0-7 days averaged a 4.42 star review compared to only 3.12 star review for delivery times of 22+ days. 13% of Olist orders took 22+ days to deliver as well. For regional delivery time the northern and northeastern states averaged a 2-3x slower delivery likely due to the rough terrain and underdeveloped logistical infrastructure. Regarding the seasonal revenue trends May-August was the highest point with \$1.63 million in revenue and 10,544 orders in August. The months following the November surge were slow with December in particular underperforming with just \$843K in revenue.

So What?:

Revenue growth alone is not the main problem for Olist. The real problem is customer retention. The massive customer loss and near zero retention explains why revenue plateaued. Growth was almost entirely driven by new customers, not ones coming back. If this pattern continues, Olist will remain limited by the amount of new customers it can attract leading to an unstable growth model compared to repeat purchases.

Currently 58% of Olist's customers are 'lost' but there is still a roughly 4.7 million revenue opportunity with the 14,207 customers classified as 'needs attention' who spend \$333 on average. This presents the opportunity for Olist to grow revenue through customer retention rather than relying on bringing in new customers.

One of the main issues Olist is facing is its cohort retention, sitting at only 0.7% or lower across every cohort far below the 20%-30% industry standard. Additionally the cohort retention rate has remained stagnant throughout the 18 months worth of data signaling that this is a structural issue for Olist rather than a quick fix.

Roughly \$2.9 million of revenue comes from four different at risk categories each with a rating below 4 stars. Given the already low customer retention rate and the overall stagnation of Olist's revenue, not addressing these issues carries the risk of declining revenue for Olist.

Delivery times were one of the main driving factors behind the customer reviews, with a drop from 4.42 stars to 3.12 stars when the delivery time takes over 22 days compared to 0-7 days. Additionally northern and northeastern regions had a 2-3x longer delivery time compared to other regions. Based on these findings delivery time is a very fixable problem, indicating logistical and operational problems compared to cohort retention which reflects a deeper overall customer experience. Because of this, addressing the delivery time issue is one of the lowest risk and highest reward actions Olist can do.

Recommendations:

1. **Create an email marketing campaign to target those 14,207 customers in the classified section of 'needs attention' offering them discounts for their next purchase.** There is 4.7 million in potential revenue so even with a realistic 15% response rate and a 20% discount per customer, Olist is looking at an additional 560K in revenue.
2. **Work with local regional fulfillment centers in northern and northeastern regions and also warn customers in those regions that deliveries can take multiple weeks.** Delivery times in northern and northeastern regions take 2-3x longer compared to other regions. By working with local regional centers and warning customers beforehand Olist can lessen delivery times while at the same time setting realistic customer expectations. This will ultimately lead to higher reviews and customer retention.
3. **Provide incentives for customers to refer new people to Olist, an example can be 20% off next order for if they bring in at least one new customer.** For example if a realistic 5% of the 12,273 loyalists and 9,335 recent customers are able to refer to one person then Olist is looking at a potential \$173K in additional revenue based on the AOV of \$159.86. The cost of this campaign would be \$34.5K to account for the 20% discount assuming the referral person had an order value of \$159.86. Beyond the direct revenue from new customers, the

referral system will also encourage existing customers to stay involved with Olist as they will have a reason to return and continue making purchases which addresses the retention problem stated earlier.

4. **Provide special package labeling/protection and make sure more experienced carriers work with items in at risk categories.** Currently \$2.9 million in revenue is generated from at-risk items with ratings below 4 stars with the largest category being Bed, Bath and Tables generating \$1 million in revenue but only rated at 3.92 stars. Even a modest improvement of how these items are shipped can increase the rating to over 4 stars and make sure that the \$2.9 million in revenue remains protected. Currently Olist is experiencing plaguing revenue so its important to protect its sources so that a drop in revenue does not occur.